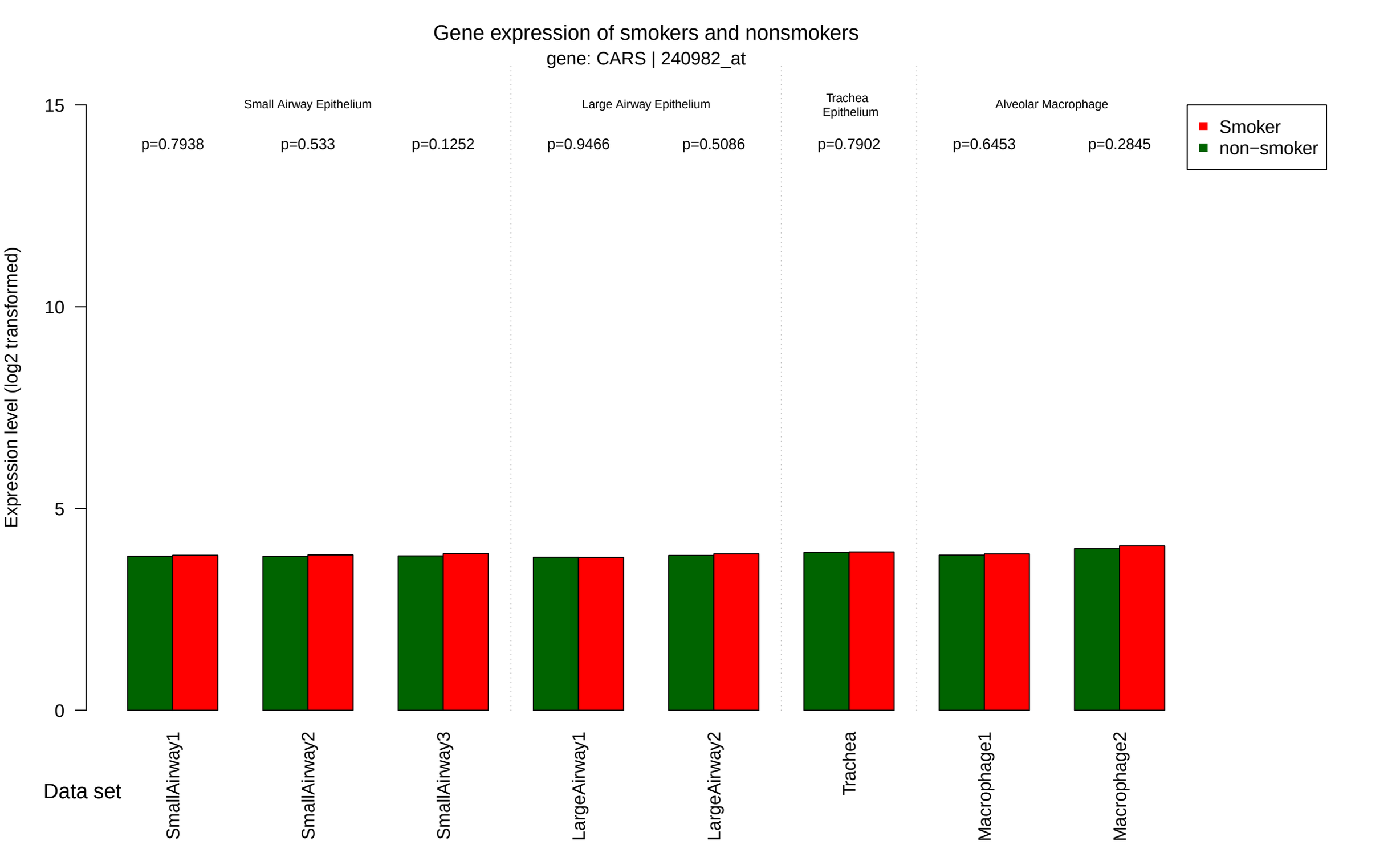
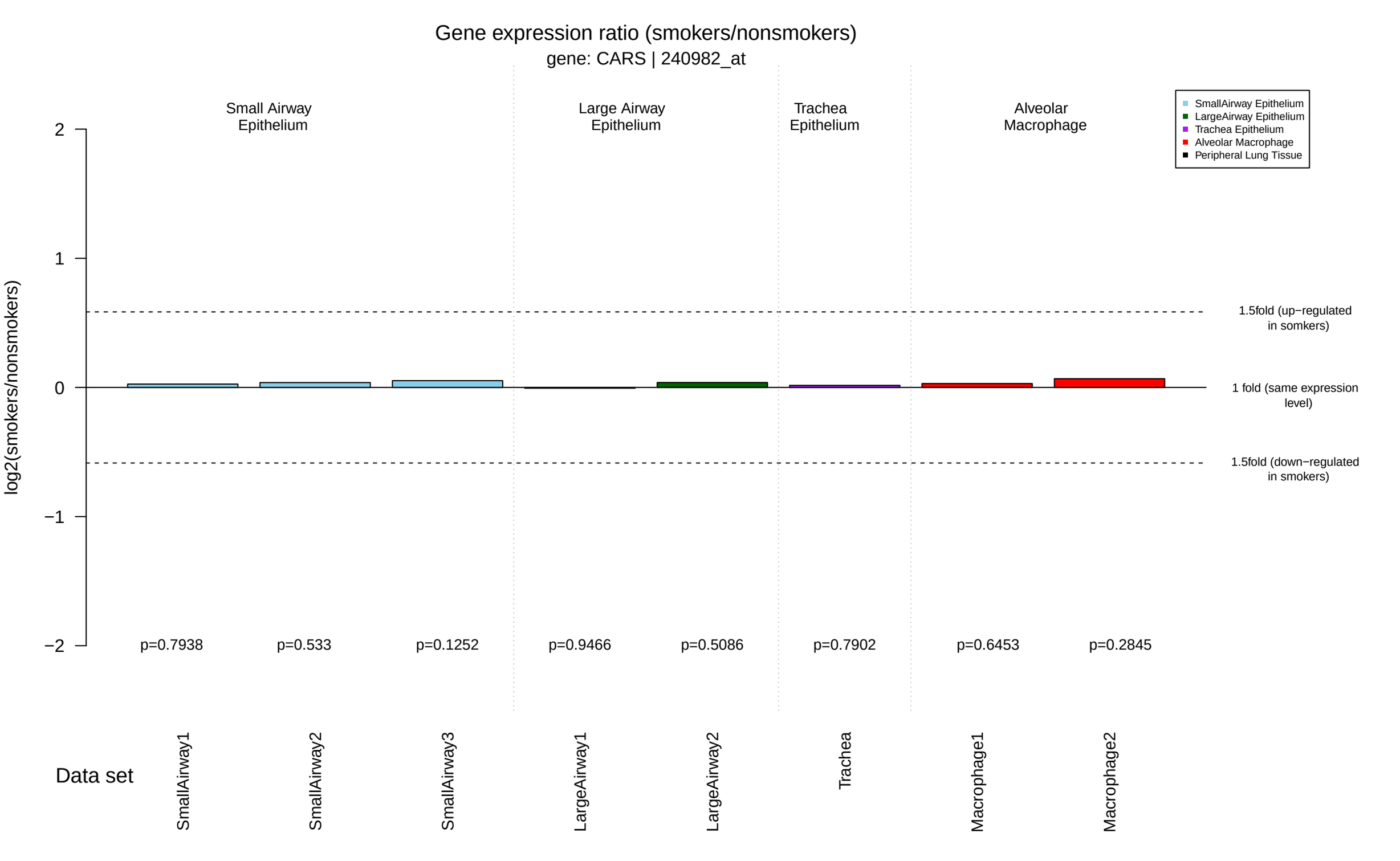
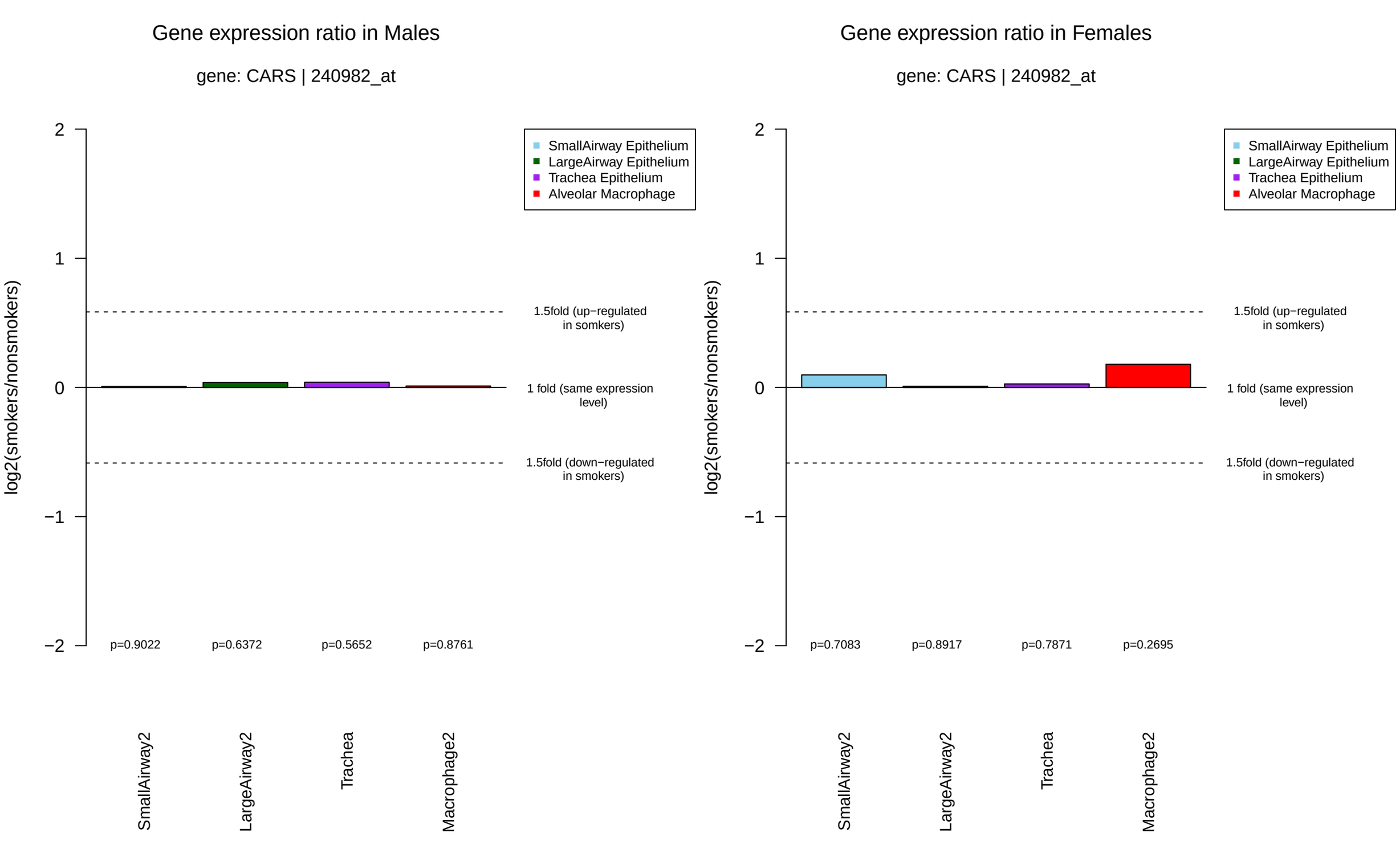
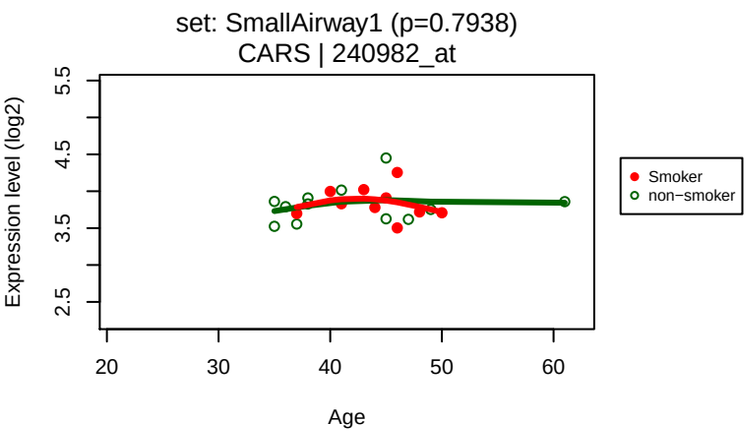
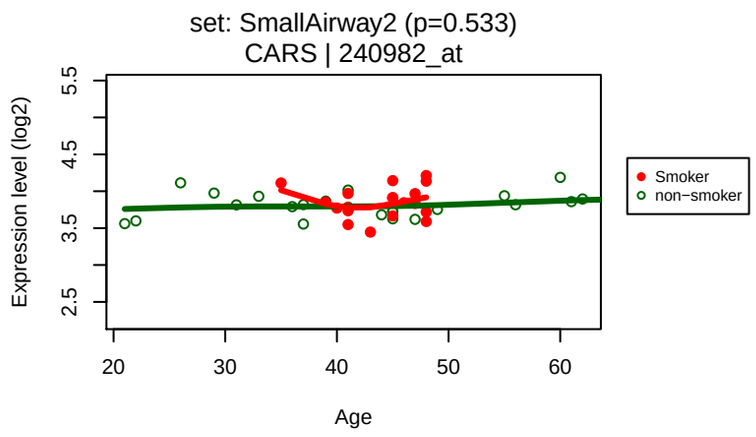
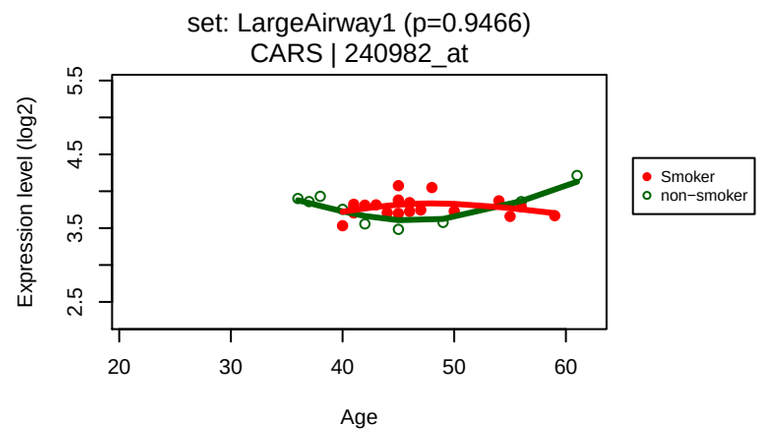

Gene expression of smokers and nonsmokers
gene: CARS | 240982_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7938
p=0.533
p=0.1252
p=0.9466
p=0.5086
p=0.7902
p=0.6453
p=0.2845
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

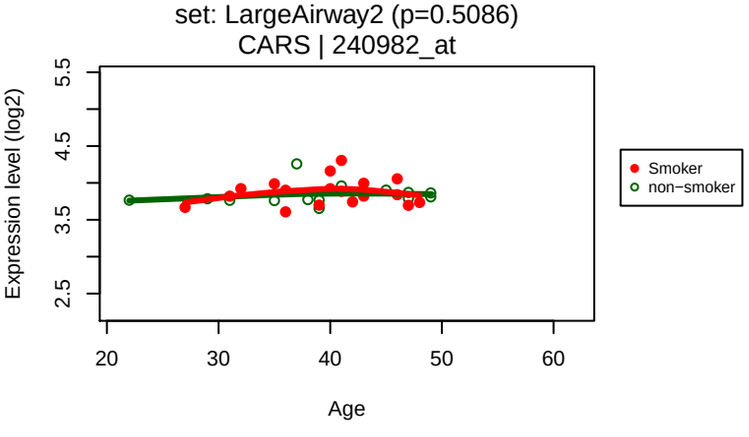

Gene expression ratio (smokers/nonsmokers)
gene: CARS | 240982_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7938
p=0.533
p=0.1252
p=0.9466
p=0.5086
p=0.7902
p=0.6453
p=0.2845
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

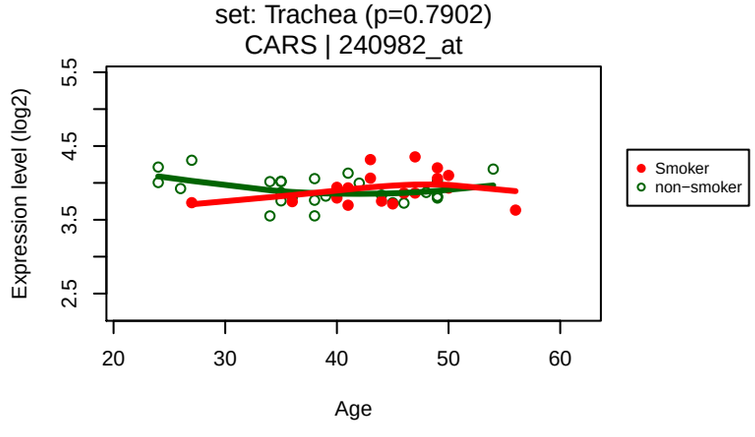

Gene expression ratio in Males
gene: CARS | 240982_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9022
p=0.6372
p=0.5652
p=0.8761
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARS | 240982_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7083
p=0.8917
p=0.7871
p=0.2695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

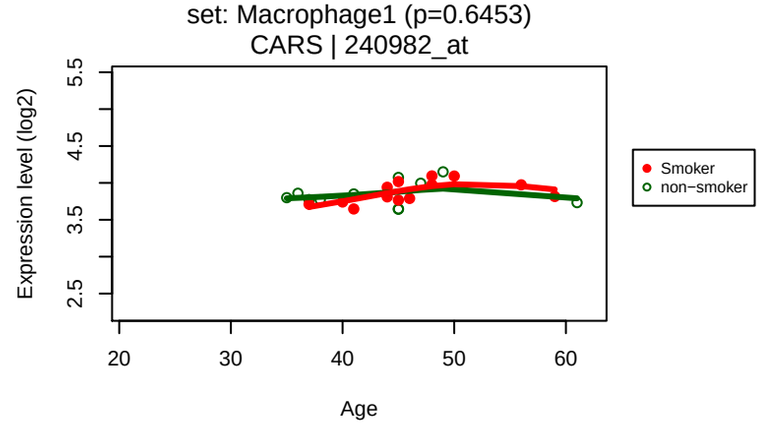

set: SmallAirway1 (p=0.7938)
CARS | 240982_at
Expression level (log2)
Age
Smoker
non−smoker

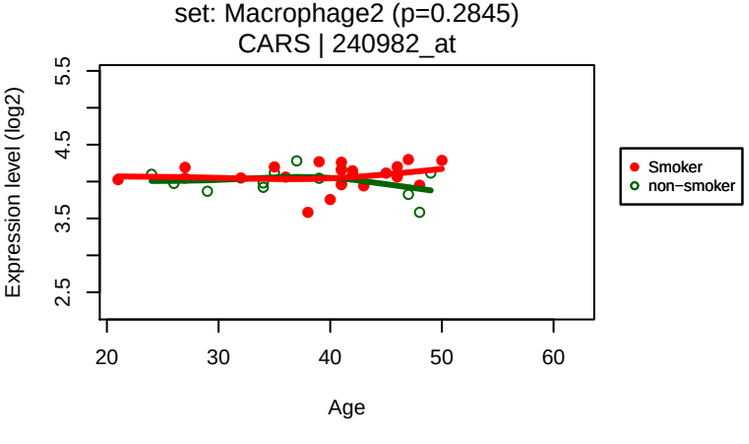

set: SmallAirway2 (p=0.533)
CARS | 240982_at
Expression level (log2)
Age
Smoker
non−smoker

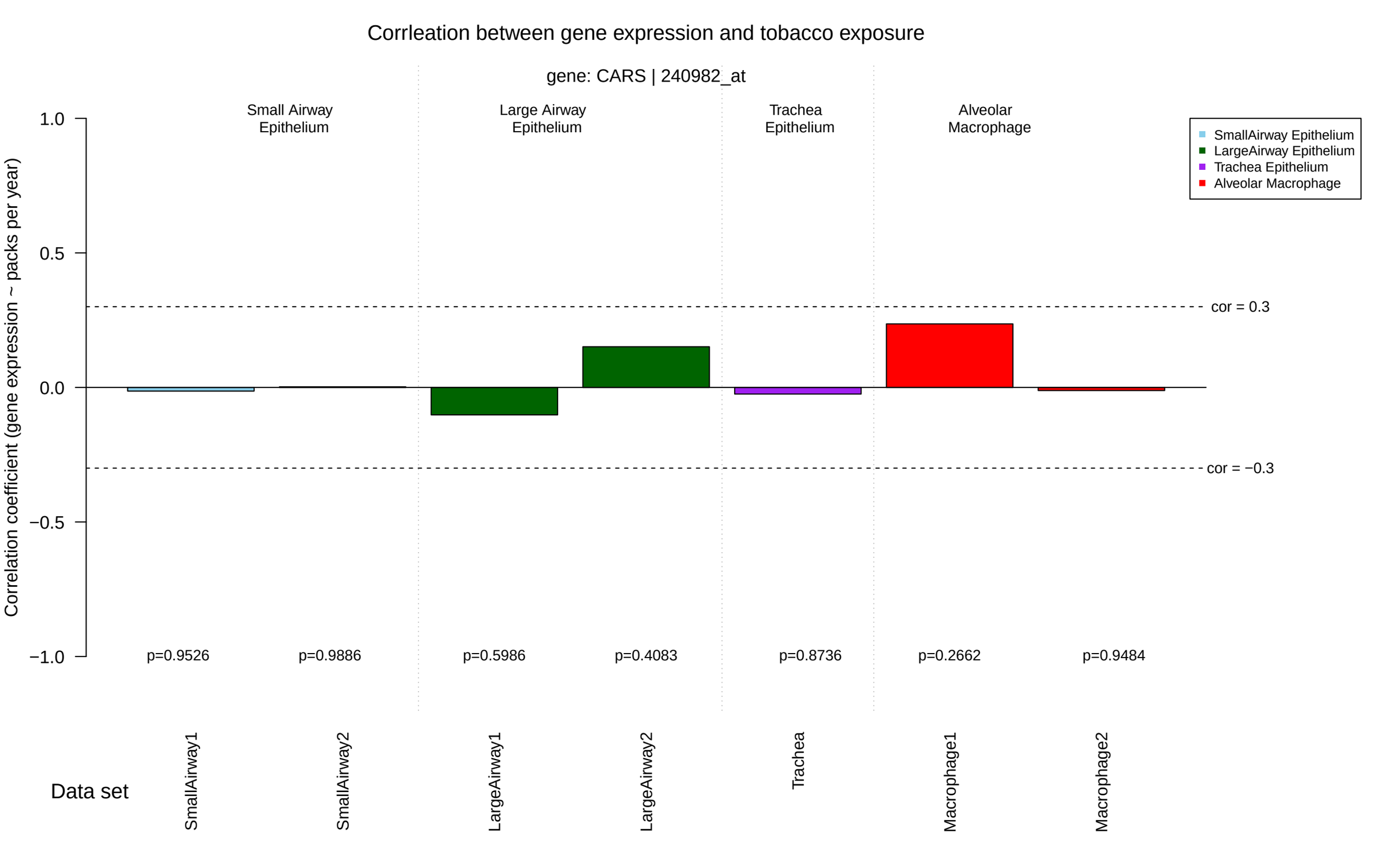

set: LargeAirway1 (p=0.9466)
CARS | 240982_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5086)
CARS | 240982_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7902)
CARS | 240982_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6453)
CARS | 240982_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2845)
CARS | 240982_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARS | 240982_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9526
p=0.9886
p=0.5986
p=0.4083
p=0.8736
p=0.2662
p=0.9484
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2